

作业纸

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$$T7.11. \therefore x(t) = \begin{cases} 1 & 3 \leq t \leq 5 \\ 0 & \text{其它} \end{cases} \quad h(t) = \begin{cases} e^{-3t} & t \geq 0 \\ 0 & \text{其它} \end{cases}$$

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = \int_3^5 e^{-3(t-\tau)} u(t-\tau)d\tau$$

$$1^\circ t < 3, \Rightarrow t - \tau < 0, \quad y(t) = 0$$

$$3^\circ 3 \leq t \leq 5, \Rightarrow u(t-\tau) = 1, \quad y(t) = \int_3^t e^{-3(t-\tau)} d\tau = \frac{1}{3}(1 - e^{-3t+9})$$

$$t \geq 5, \quad y(t) = \int_3^5 e^{-3(t-\tau)} d\tau = \frac{1}{3}(e^{-3t+3} - e^{-3t+5})$$

$$\therefore y = \begin{cases} 0 & t < 3 \\ \frac{1}{3}(1 - e^{-3t+9}) & 3 \leq t < 5 \\ \frac{1}{3}(e^{-3t+3} - e^{-3t+5}) & t \geq 5 \end{cases}$$

$$(2) \frac{dx(t)}{dt} = \delta(t-3) - \delta(t+5)$$

$$\therefore g(t) = [\delta(t-3) - \delta(t+5)] * e^{-3t}u(t)$$

$$= e^{-3t-3}u(t-3) - e^{-3t+5}u(t+5)$$

$$(3) g(t) = \begin{cases} 0 & t < 3 \\ e^{-3t+3} & 3 \leq t < 5 \\ e^{-3t+3} - e^{-3t+5} & t \geq 5 \end{cases}$$

$$y(t) = \begin{cases} 0 & t < 3 \\ e^{-3t+3} & 3 \leq t < 5 \\ e^{-3t+3} - e^{-3t+5} & t \geq 5 \end{cases}$$

$$\therefore g(t) = y(t)$$

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T2.17. $\frac{dy(t)}{dt} + 4y(t) = x(t)$.

$$sY + 4Y = X \Rightarrow H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+4}$$

输入 $x(t) = e^{-(1+3j)t} u(t)$ 拉普拉斯变换为

$$X(s) = \frac{1}{s - (-1+3j)}$$

$$\therefore Y(s) = \frac{1}{(s+4)(s+1-3j)} = \frac{A}{s+4} + \frac{B}{s+1-3j}$$

即: $1 = A(s+1-3j) + B(s+4)$. 令 $s = -4 \Rightarrow 1 = A(-3-3j) \Rightarrow A = \frac{j-1}{6}$

同理, $B = \frac{1-j}{6}$

$$\therefore y(t) = (Ae^{-4t} + Be^{-(1+3j)t}) u(t) = \frac{1}{6} (e^{-(1+3j)t} - e^{-4t})(1-j) u(t)$$

12) $x(t) = e^{-t} \cos(3t) u(t) = \operatorname{Re} \{ e^{(3j)t - t} u(t) \} = \operatorname{Re} \{ e^{(-1+3j)t} u(t) \}$

$y'(t) + 4y(t) = x(t)$, 由(1)知, $x(t) = \operatorname{Re} \{ X_{10}(t) \}$

$$\therefore y(t) = \operatorname{Re} \{ y_{10}(t) \} = \operatorname{Re} \left\{ \frac{1}{6} e^{(-1+3j)t} - e^{-4t} \right\} (1-j) u(t)$$

$$= \operatorname{Re} \{ (1-j) (e^{-t} (\cos 3t + j \sin 3t) - e^{-4t}) \} = \frac{1}{6} [e^{-t} (\cos 3t + \sin 3t) - e^{-4t}] u(t)$$

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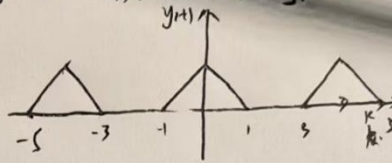
$$T=2, h(t) = \begin{cases} t+1 & (-1 \leq t \leq 0) \\ 1-t & (0 \leq t \leq 1) \\ 0 & \text{其它} \end{cases}$$

$$y(t) = \gamma(t) * h(t) = \sum_{k=-\infty}^{+\infty} h(t-kT)$$

* $h(t)$ 非零区间为 $[-1, 1]$, $\Rightarrow y(t)$ 非零为 $[kT-1, kT+1]$.

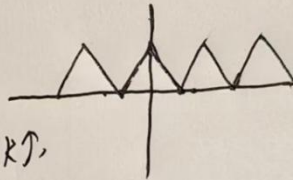
$$1^\circ T=4 \Rightarrow y(t) = \sum_{k=-\infty}^{+\infty} h(t-4k)$$

$\therefore y(t)$ 非零为 $[4k-1, 4k+1]$.



2. $T=2$, $y(t)$ 非零为 $[2k-1, 2k+1]$

相邻脉冲在端点相接.

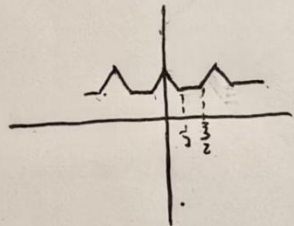


3. $T = \frac{3}{2} \Rightarrow y(t)$ 非零为 $[\frac{3}{2}k-1, \frac{3}{2}k+1]$, 当 k 奇, 脉冲有相交.

$$\text{对重叠处} \Rightarrow \frac{3}{2}(k+1)-1 = \frac{3}{2}k+1.$$

在 $[\frac{3}{2}k+\frac{1}{2}, \frac{3}{2}(k+1)]$ 处重叠.

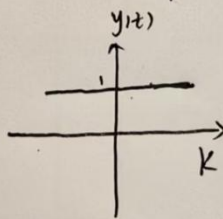
重叠处为恒定值 $\frac{1}{2}$.



4. $T=1$, $\Rightarrow y(t)$ 非零为 $[k-1, k+1]$.

脉冲中有相接, 相邻为 $[k, k+1]$.

$\therefore$ 重叠相邻恒为 1.



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$$T_{24}. Y = \sum_{k \in [3, 8]} x[k] h[n-k].$$

$$k \in [3, 8], x[k] = 1.$$

$$n-k \in [4, 15] \Rightarrow k \in [n-15, n-4], h[n-k] = 1.$$

$$\bar{0} x[k] \text{ 与 } h[n-k] \text{ 交集非空} \Rightarrow \begin{cases} n-15 \leq 8 \\ n-4 \geq 3 \end{cases} \Rightarrow 7 \leq n \leq 23.$$

$$1^{\circ} \cdot \begin{cases} 3 \leq k \leq 8 \\ n-4 \leq k \leq n-3 \end{cases} \Rightarrow \begin{cases} n-15 \leq 3 \\ n-4 \geq 8 \end{cases} \Rightarrow 7 \leq n \leq 12, 3 \leq k \leq n-4.$$

$$y[n] = n-4-3+1 = n-6.$$

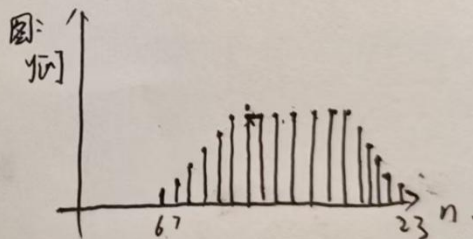
$$2^{\circ} \cdot 3 \leq k \leq 8 \Rightarrow \begin{cases} n-15 \leq 3 \\ n-4 \geq 8 \end{cases} \Rightarrow 12 \leq n \leq 18, 3 \leq k \leq 8.$$

$$y[n] = 8-3+1 = 6.$$

$$3^{\circ} \cdot n-15 \leq k \leq 8 \Rightarrow \begin{cases} n-15 \geq 3 \\ n-4 \leq 8 \end{cases} \Rightarrow 19 \leq n \leq 23.$$

$$y[n] = 8-n+15+1 = 24-n.$$

$$\therefore y[n] = \begin{cases} n-6 & 7 \leq n < 12 \\ 6 & 12 \leq n \leq 18 \\ 24-n & 19 \leq n \leq 23 \\ 0 & \text{其它} \end{cases}$$



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